A General Cost Function for Keyboard Layouts

Sensitivity, Calibration and Validity

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Working draft skeleton · no results yet · bracketed TODOs are placeholders
Focus paper 2. Topics T1, T5, T8, T11. Status: scorer, profiles, judges, the pairwise fit, weight sweeps (`gecko sweep`, `compare --robust`) and the E01 timing model exist; a user study does not.

Abstract

Layout tools disagree because their cost functions disagree. We define a general cost function: a weighted sum of geometry-derived metrics over mono, bigram, skipgram and trigram statistics plus a few non-linear balance terms, parameterized by a board, a corpus and a profile, and we ask what makes such a function good. We test it three ways: **sensitivity** (how far the optimum moves with the weights), **validity** (agreement with independent judges: Engram comfort, keystroke timing, community preference) and **robustness** (transfer across corpora and boards). [TODO: key findings.] We give a recipe for a defensible default profile and report where the evidence is weak.

1. Research questions

1. RQ1 *Expressiveness*: which metric families are needed (ablation: drop a family, measure what changes in rankings and optima)?
2. RQ2 *Sensitivity*: how many optimum families does the plausible weight space contain; which weights decide between them (Sobol indices)?
3. RQ3 *Validity*: how well does the score predict (a) IKI from 136M keystrokes, (b) Engram pairwise comfort, (c) community likes, (d) a small user study?
4. RQ4 *Goodhart*: does more search effort widen the gap between proxy and judge?
5. RQ5 *Transfer*: does a profile fitted on one corpus/board hold on others (english, shai, reddit, mt-10k, french, code; ANSI-30, Corne)?

2. Hypotheses (not results)

H1 a few families explain most ranking changes. H2 two or three weights dominate the optimum. H3 geometry-based metrics transfer better than trigram metrics. H4 learned pair-feature weights beat hand-set weights on held-out typists, but only modestly. H5 the proxy-vs-judge gap grows with search effort unless the judge is part of the objective.

3. Method

- Definition: `context/objective.md` (folding, spread-normalized weights, non-linear terms).
- Sensitivity: Latin-hypercube over log-scaled spread-normalized weights; archive once with `gecko explore`, re-rank under each vector (`gecko atlas -p`); embedding by finger distance.
- Validity: IKI regression on geometric pair features, split by typist; pairwise logistic fit (`gecko profile fit --target likes`) with bootstrap CIs; judges never used to train and test the same claim.
- Goodhart: budget grid 1 s … 10 min; optimize profile A, score on B, C and E05.
- Pre-register the analysis (cluster definition, cutoffs) before running.

4. Existing evidence

Timing validity, comfort agreement, weight robustness and a 600-run weight sweep: the experiment report `docs/papers/cost-validity/` (`research/experiments/2026-10-22-cost-function-validit`
The default profile was retuned from it on 2026-10-22; a sweep of the new default is in `bench/sweep-2026-10-22/`, and SFS timing by context in `research/experiments/2026-10-22-skipgram-timing.md`. Held-out pairwise accuracy 53.78 % (learned) vs 51.96 % (default) vs 49.40 % (Engram E05) (`research/experiments/2026-10-21-profile-ranking.md`; no CIs yet). An earlier optimum (8.396) ranked last on E05 after the profile changed; B11 at weight 1 re-sorts Halmak and Colemak (`context/objective.md`).

5. Figures

F1 metric-family ablation · F2 optimum embedding by weight vector · F3 Sobol bars · F4 regret matrix · F5 validity: predicted vs observed (IKI heatmap by finger

pair; accuracy with CIs) · F6 proxy-vs-judge along search · F7 transfer matrix · F8 consensus keyboard.

6. Threats

Timing data is from QWERTY typists (learn geometric, never per-letter costs); likes measure popularity; E05 comes from one study; hand model parameters from literature; no user study yet.

7. Needed

Bootstrap CIs on the ranking fits; Sobol indices over the weights; ideally a 10–20 person study (timing and pairwise comfort, including SFS).